

VEXAR Fleet Analytics

Technical Report — Data Scientist Intern Assignment

VexarDrive Technologies

Two-Wheeler Delivery / Rider Fleet Analytics

1. Executive Summary

This project analyzes one week of two-wheeler fleet data covering **30 drivers, 30 vehicles, 450 trips, and 12,987 minute-level telemetry observations**. It produces two interactive dashboards: a Driver Behaviour Dashboard and a Vehicle Health Status Dashboard.

Driver behaviour is evaluated using speeding, harsh acceleration, harsh braking, lateral movement, and gyroscope activity. Vehicle health is evaluated using vibration anomalies, gyroscope anomalies, and Z-axis variability.

The scoring framework is deliberately explainable and heuristic because the supplied dataset contains no labelled accident, incident, breakdown, or mechanical-failure outcomes. Component event rates are normalized for telemetry volume and converted into fleet-relative percentile scores. Therefore, the scores are operational prioritization indicators within this fleet, not absolute accident probabilities or mechanical diagnoses.

2. Problem Statement and Objectives

The objective is to identify relatively risky versus safe driving patterns per driver and vehicles showing unusual sensor signatures that may warrant maintenance review. The project also documents the methodology, assumptions, and limitations behind each major metric.

Key objectives:

- Identify and rank driver behaviour using explainable sensor-based event rates.
- Identify and prioritize vehicles with unusual sensor signatures.
- Provide transparent methodology and assumptions for dashboard metrics.
- Propose additional applications for the same fleet telemetry.

3. Dataset and Data Model

Sheet	Purpose
Drivers	Driver master data.
Vehicles	Vehicle master data and maintenance context.
Trips	Trip-level duration, distance and speed summaries.
Telemetry	Minute-level GPS, accelerometer and gyroscope readings.

Join relationships: Telemetry.Trip_ID → Trips.Trip_ID; Trips.Driver_ID → Drivers.Driver_ID; Trips.Vehicle_ID → Vehicles.Vehicle_ID.

The final joined telemetry dataset contains 12,987 observations.

4. Data Validation and Quality Checks

Check	Result
Unique drivers	30
Unique vehicles	30
Unique trips	450
Trips per driver	15 for each driver

Missing values	0 in validated columns
Duplicate records	0
Invalid driver/vehicle/trip relationships	0
Telemetry records	12,987

Sensor plausibility checks covered speed above 80 km/h, negative speed, absolute X/Y acceleration above 1g, negative Z acceleration, and gyroscope magnitude above 60 dps. No observations exceeded these sanity limits. Extreme observations were retained rather than automatically removed because they may represent genuine riding events.

5. Exploratory Data Analysis

Speed had a mean of 24.19 km/h, median 24.90 km/h, 95th percentile 41.80 km/h, 99th percentile 51.30 km/h, and maximum 73.60 km/h. Accel_X had standard deviation 0.156g, Accel_Y 0.087g, and Accel_Z was centered near 1g with mean 1.007g and standard deviation 0.076g. Gyroscope magnitude had median 3.18 dps and 95th percentile 7.58 dps.

Threshold testing found speed above 50 km/h in 167 observations (1.29%), harsh acceleration above +0.35g in 423 (3.26%), harsh braking below -0.35g in 389 (3.00%), lateral acceleration magnitude above 0.25g in 385 (2.96%), and gyroscope magnitude above 10 dps in 579 (4.46%).

6. Driver Behaviour Methodology

Indicator	Threshold	Weight
Speeding	> 50 km/h	25%
Harsh acceleration	Accel_X > +0.35g	20%
Harsh braking	Accel_X < -0.35g	25%
Lateral movement	Accel_Y > 0.25g	15%
Gyroscope activity	Magnitude > 10 dps	15%

For each driver, event rate = qualifying telemetry observations / total telemetry observations × 100. This normalizes differences in telemetry volume.

Weight rationale: weights are operational heuristics, not statistically learned coefficients. Speeding and harsh braking receive the highest weights because they are treated as the most direct high-risk longitudinal behaviours available. Harsh acceleration is slightly lower because it can be aggressive without necessarily being unsafe. Lateral acceleration and gyroscope activity receive lower weights because elevated values can occur during legitimate cornering, turning, road-surface changes, or normal manoeuvres.

Component rates are converted into fleet-relative percentile scores and combined into a 0–100 Driver Risk Score. This is a fleet-relative prioritization indicator, not an absolute accident-risk probability.

7. Driver Results

Rank	Driver	Score	Category
1	D06 — Bhavani Raj	91.67	High Risk
2	D03 — Vignesh Murugan	90.83	High Risk
3	D19 — Senthil Pillai	90.00	High Risk
4	D23 — Kavya Pillai	88.17	High Risk
5	D14 — Rajesh Subramaniam	88.00	High Risk

The generated driver analysis classified 11 drivers as High Risk, 10 as Moderate, and 9 as Safe. Because percentile ranking is fleet-relative, this near-even category split is a property of the scoring approach and should not be interpreted as evidence that exactly one-third of the fleet is objectively unsafe.

8. Vehicle Health Methodology

Indicator	Definition	Weight
Vibration anomaly rate	$ Accel_Z - 1g > 0.15g$	45%
Gyroscope anomaly rate	Magnitude > 10 dps	30%
Z-axis variability	Std. deviation of Z-axis deviation	25%

The vehicle-health weights are heuristic because no labelled mechanical failures are available. Vibration anomaly receives the highest weight because repeated abnormal vertical acceleration is a useful sensor-based indicator of irregular ride behaviour. Gyroscope anomaly receives a secondary weight because unusual rotational behaviour may indicate irregular dynamics, although it can also arise from normal manoeuvres. Z-axis variability captures consistency of vertical sensor behaviour.

Vehicle age, odometer, and service recency are deliberately shown as maintenance context rather than direct evidence of mechanical failure. The health score remains focused on sensor-based anomaly evidence.

Vibration anomaly rate and Z-axis variability are both derived from vertical acceleration and therefore represent related rather than completely independent evidence.

9. Vehicle Results

The vehicle analysis classified 10 vehicles as Maintenance Review, 12 as Moderate, and 8 as Safe under the current fleet-relative scoring scheme. The highest-scoring vehicle in the generated output was V02 with a Health Risk Score of 100.00.

A Maintenance Review classification is a candidate for inspection, not a confirmed mechanical failure. Operational teams should combine sensor signals with service history, physical inspection, and manufacturer guidance.

10. Dashboard Design

The Streamlit application provides an Executive Overview, Driver Behaviour Dashboard, and Vehicle Health Dashboard. The overview presents fleet KPIs, observed driver and vehicle priorities, score interpretation, methodology, and future applications. Detailed dashboards provide entity-level scores and supporting metrics.

Threshold-context percentages shown in the vehicle methodology are calculated from the loaded master telemetry dataset rather than hardcoded result values.

11. Assumptions

- The supplied telemetry is representative of the one-week fleet period.
- Selected thresholds are analytical, not regulatory or manufacturer limits.
- IMU axes are interpreted according to supplied field names and units.
- Percentile scores provide fleet-relative prioritization because labelled outcomes are unavailable.
- Extreme observations are retained after plausibility screening unless they violate predefined sanity checks.
- Maintenance context variables are not treated as direct evidence of mechanical failure.

12. Limitations

- No accident, incident, breakdown, or mechanical-failure labels are available.
- Scores are heuristic and have not been validated against real-world outcomes.
- Percentile scoring is relative to the supplied fleet and can produce a roughly proportional category split by construction.
- Telemetry counts vary across entities, so low-volume rates may be less stable.
- Some sensor components are correlated, including the two Z-axis-derived vehicle-health measures.
- No statistical shrinkage, confidence intervals, correlation adjustment, or sensitivity analysis is currently applied.
- The dashboard should not be used as an accident predictor or mechanical-failure diagnosis.

13. Additional Applications

Predictive maintenance: combine telemetry anomalies with historical maintenance and breakdown outcomes.
Driver safety coaching: identify recurring behaviour patterns for targeted coaching.
Accident / incident detection: flag extreme combinations for investigation after validation.
Route risk analysis: identify road segments associated with repeated harsh manoeuvres.
Trip-time prediction: use distance, speed, route, and duration features for ETA prediction.
Fleet utilization optimization: improve vehicle allocation using trip demand and usage.
Sensor anomaly detection: identify calibration issues or sensor degradation.
Driver performance benchmarking: track behaviour over longer periods.
Maintenance prioritization: combine sensor signals with service history.

14. Future Improvements

If labelled accident and maintenance outcomes become available, the scoring framework could be evaluated against real-world outcomes and supplemented or replaced by supervised predictive models. Historical maintenance records could calibrate vehicle-health thresholds. Correlation and sensitivity analysis could assess feature redundancy and weight stability. Confidence intervals or statistical shrinkage could account for different telemetry volumes.

15. Conclusion

The completed solution provides an explainable and reproducible workflow from raw workbook validation through joined telemetry analysis, driver scoring, vehicle-health scoring, and interactive visualization. Its principal strength is transparency: thresholds, weights, assumptions, and limitations are explicitly documented. The dashboards should be used as operational prioritization tools and recalibrated when historical safety and maintenance outcomes become available.